EDA Steps for Trading Data

Student instructions using trades_orderbook_STUDENT_dirty.csv (synthetic BTCUSDT data)

Note: this is data preparation and feature engineering, not model fine-tuning.

1. Load and inspect

- Read the CSV with pandas, then run **head()**, **shape**, **info()**, and **dtypes**.
- Note the column types and the row count.

2. Summarize

- Run **describe()** and **value_counts()** on side and symbol.
- Check min, max, mean, and median for price and quantity.

3. Find the problems

- Count missing values, duplicates, mixed timestamp formats, negative or zero values, and crossed books (bid above ask).

4. Clean

- Drop duplicates by trade_id.
- Standardize side and symbol.
- Convert the price text to numbers.
- Parse all timestamps into one format.
- Fix or drop bad rows, and write down why for each decision.

5. Handle outliers

- Find extreme prices with a rolling median or the IQR method, then remove or correct them.

6. Engineer features

- spread = ask_1 - bid_1
- mid_price = (ask_1 + bid_1) / 2
- order_imbalance = (bid_qty_1 - ask_qty_1) / (bid_qty_1 + ask_qty_1)
- Returns and rolling volatility
- Optionally RSI and moving averages

7. Resample

- Convert the ticks into 1-minute OHLCV candles, sorted by time.

8. Visualize

- Price line chart
- Volume histogram
- Spread distribution
- Buy vs sell counts
- Correlation heatmap of the features

9. Prepare for a model

- Define a target, such as "price up or down in the next 5 minutes".
- Split by time, not randomly (for example the first 70% for training, the last 30% for testing).
- Scale the features if needed.
- Save the cleaned file.

For the 3Commas demo

Cleaned data, then features, then a simple model (even a rule like "buy when order imbalance is above 0.3"), then a backtest, then buy/sell signals sent to 3Commas by webhook, tested in paper trading first.

Rules for students

- Never shuffle time-series data. Split by time only.
- Avoid look-ahead bias, meaning no feature may use future data.
- Document every cleaning decision.

Educational material, not financial advice. Data is synthetic.